STT 465 Homework 3 — Answer Key

Normal Models, Prior Choice, and Unknown Variance

Notation: Gamma uses the rate parameterization, Gamma(a, b) has density $\propto \theta^{a-1}e^{-b\theta}$ and mean a/b. IG(a, b) has density $\propto (\sigma^2)^{-a-1}e^{-b/\sigma^2}$ and mean $b/(a-1)$.

Question 1: Normal Model with Known Variance

(a) The likelihood kernel in μ is

$$f(x \mid \mu) \propto \exp\Big\{ -\tfrac{1}{2\sigma^2}\textstyle\sum_i (x_i-\mu)^2 \Big\} \propto \exp\Big\{ -\tfrac{n}{2\sigma^2}(\mu^2 - 2\mu\bar{x}) \Big\},$$

using $\sum(x_i-\mu)^2 = \sum(x_i-\bar{x})^2 + n(\bar{x}-\mu)^2$ and dropping terms free of μ. The prior kernel is $\pi(\mu) \propto \exp\{-\frac{1}{2\tau_0^2}(\mu^2 - 2\mu\mu_0)\}$. Multiplying,

$$\pi(\mu \mid x) \propto \exp\Big\{ -\tfrac{1}{2}\Big[\Big(\tfrac{1}{\tau_0^2} + \tfrac{n}{\sigma^2}\Big)\mu^2 - 2\mu\Big(\tfrac{\mu_0}{\tau_0^2} + \tfrac{n\bar{x}}{\sigma^2}\Big) \Big] \Big\}.$$

This is the kernel of a normal: $\exp\{-\frac{1}{2\tau_n^2}(\mu^2 - 2\mu\mu_n)\}$ with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \tau_n^2\Big(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}\Big).$$

Answer. $\mu \mid x \sim N(\mu_n, \tau_n^2)$. The *posterior precision* is $1/\tau_n^2 = 1/\tau_0^2 + n/\sigma^2$: prior precision plus data precision.

(b) Distributing τ_n^2,

$$\mu_n = \underbrace{\frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2}}_{w}\,\bar{x} + \underbrace{\frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2}}_{1-w}\,\mu_0.$$

Answer. $w = \dfrac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2} = \dfrac{n\tau_0^2}{n\tau_0^2 + \sigma^2} \in (0,1)$. The weight on $\bar{x}$ is the data's share of the total precision. A large n, a small σ^2 or a vague prior (large τ_0^2) push $w \to 1$, so the posterior mean follows the data. A tight prior (small τ_0^2) pushes $w \to 0$, so the posterior mean stays near μ_0.

(c) Write $X_f = \mu + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$ independent of μ given x. Given x, $\mu \sim N(\mu_n, \tau_n^2)$, so X_f is a sum of independent normals. Equivalently, $p(x_f \mid x) = \int N(x_f; \mu, \sigma^2)\, N(\mu; \mu_n, \tau_n^2)\, d\mu$, and completing the square in μ leaves a normal kernel in x_f. Its moments are $E[X_f \mid x] = E[E(X_f \mid \mu)] = \mu_n$ and $\mathrm{Var}(X_f \mid x) = E[\sigma^2] + \mathrm{Var}(\mu \mid x) = \sigma^2 + \tau_n^2$.

Answer. $X_f \mid x \sim N(\mu_n,\ \sigma^2 + \tau_n^2)$. The variance combines sampling noise (σ^2) and remaining uncertainty about μ (τ_n^2).

(d) Here $n/\sigma^2 = 16/4 = 4$ and $1/\tau_0^2 = 1$.

$$\tau_n^2 = (1+4)^{-1} = 0.2, \qquad \mu_n = 0.2\,(4/1 + 4 \cdot 5.4) = 0.2(4 + 21.6) = 5.12.$$

Answer. $\mu \mid x \sim N(5.12,\ 0.2)$; posterior mean 5.12, variance 0.2. Predictive: $X_f \mid x \sim N(5.12,\ 4.2)$. (Here $w = 4/5 = 0.8$.)

(e) With $\tau_0^2 = 1000^2$, prior precision $= 10^{-6}$:

$$\tau_n^2 = (10^{-6} + 4)^{-1} \approx 0.25, \qquad \mu_n = \tau_n^2(0 + 21.6) = \frac{21.6}{4.000001} \approx 5.39999 \approx 5.4.$$

Answer. $\mu \mid x \approx N(5.40,\ 0.25)$, and $X_f \mid x \approx N(5.40,\ 4.25)$. The posterior means differ by only $5.40 - 5.12 = 0.28$. The difference is small because the data carry four times the information of the informative prior (precision 4 vs. 1, so $w = 0.8$ already). Even the informative prior has only 20% weight, and the diffuse prior has essentially none ($w \approx 1$, so $\mu_n \approx \bar{x}$). Switching priors therefore moves the estimate only a short distance between $\bar{x} = 5.4$ and the prior mean 4.

Question 2: Prior Choice and Reparameterization

(a) Center at $\mu_0 = 30$. The 95% range 10 to 50 is $\mu_0 \pm 20$, and $20 = 2\tau_0$.

Answer. $\mu_0 = 30,\ \tau_0 = 10$, so $\mu \sim N(30,\ 100)$.

(b) $\int_{-\infty}^{\infty} 1\, d\mu = \infty$, so $\pi(\mu) \propto 1$ cannot be normalized to a probability density. That makes it improper. The posterior is

$$\pi(\mu \mid x) \propto 1 \cdot \exp\Big\{ -\tfrac{1}{2\sigma^2} \textstyle\sum (x_i - \mu)^2 \Big\} \propto \exp\Big\{ -\tfrac{n}{2\sigma^2}(\mu - \bar{x})^2 \Big\}.$$

For any $n \geq 1$, $\int \exp\{-\frac{n}{2\sigma^2}(\mu - \bar{x})^2\} d\mu = \sqrt{2\pi\sigma^2/n} < \infty$, so the posterior normalizes.

Answer. $\mu \mid x \sim N(\bar{x},\ \sigma^2/n)$, which is proper for every $n \geq 1$. It is the $\tau_0^2 \to \infty$ limit of Question 1.

(c) $\pi(\sigma^2) = \frac{1}{10}$ on $(0, 10)$. Let $\tilde{\sigma} = 1/\sigma^2$, so $\sigma^2 = g(\tilde{\sigma}) = 1/\tilde{\sigma}$ and $|g'(\tilde{\sigma})| = \tilde{\sigma}^{-2}$. Because $\sigma^2 \in (0, 10)$, $\tilde{\sigma} \in (1/10, \infty)$. Then

$$\pi(\tilde{\sigma}) = \pi_{\sigma^2}(1/\tilde{\sigma}) \left| \frac{d\sigma^2}{d\tilde{\sigma}} \right| = \frac{1}{10} \cdot \frac{1}{\tilde{\sigma}^2}.$$

Answer. $\pi(\tilde{\sigma}) = \dfrac{1}{10\,\tilde{\sigma}^2}$ for $\tilde{\sigma} > 0.1$ (zero otherwise). Check: $\int_{0.1}^{\infty} \frac{1}{10\tilde{\sigma}^2} d\tilde{\sigma} = \frac{1}{10} \cdot 10 = 1$. A "flat" prior on σ^2 is far from flat on the precision scale.

(d) Take $\pi(\tilde{\sigma}) \propto 1$ and map to σ^2: $\tilde{\sigma} = 1/\sigma^2$, $|d\tilde{\sigma}/d\sigma^2| = (\sigma^2)^{-2}$, so

$$\pi(\sigma^2) \propto 1 \cdot (\sigma^2)^{-2}.$$

Answer. A flat prior on $\tilde{\sigma}$ implies $\pi(\sigma^2) \propto (\sigma^2)^{-2}$. That is not constant, so it is not a flat prior on σ^2; it piles mass near $\sigma^2 = 0$. Because the Jacobian reshapes any density under a nonlinear transformation, "flat" (and so "non-informative") is defined only relative to a chosen parameterization. Being uniform on one scale means being informative on another.

(e) $\pi_J(\tilde\sigma) \propto \tilde\sigma^{-1}$. With $\tilde\sigma = 1/\sigma^2$ and $|d\tilde\sigma/d\sigma^2| = (\sigma^2)^{-2}$,

$$\pi(\sigma^2) \propto (1/\sigma^2)^{-1}(\sigma^2)^{-2} = \sigma^2 \cdot (\sigma^2)^{-2} = (\sigma^2)^{-1}.$$

Answer. $\pi(\sigma^2) \propto 1/\sigma^2$. Deriving Jeffreys' prior directly on σ^2 gives the same answer: $I(\sigma^2) = \frac{n}{2\sigma^4}$, so $\sqrt{I(\sigma^2)} \propto 1/\sigma^2$. Jeffreys' rule $\pi \propto \sqrt{I(\theta)}$ is transformation-invariant: transforming the Jeffreys prior on $\tilde\sigma$ gives exactly the Jeffreys prior on σ^2. Flat priors lack this property, as (c)–(d) showed.

Question 3: Normal Model with Known Mean

(a) With $\sigma^2 = 1/\tilde\sigma$, each density is $(\tilde\sigma/2\pi)^{1/2}\exp\{-\frac{\tilde\sigma}{2}(x_i-\mu)^2\}$, so

$$\begin{aligned} f(x \mid \tilde\sigma) &= \prod_{i=1}^{n}\Big(\frac{\tilde\sigma}{2\pi}\Big)^{1/2} e^{-\frac{\tilde\sigma}{2}(x_i-\mu)^2} = (2\pi)^{-n/2}\,\tilde\sigma^{\,n/2}\exp\Big\{-\frac{\tilde\sigma}{2}SS_\mu\Big\} \\ &\propto \tilde\sigma^{\,n/2}\exp\Big\{-\frac{\tilde\sigma}{2}SS_\mu\Big\}. \qquad \checkmark \end{aligned}$$

(b) With $\pi(\tilde\sigma) \propto \tilde\sigma^{a_0-1}e^{-b_0\tilde\sigma}$,

$$\pi(\tilde\sigma \mid x) \propto \tilde\sigma^{a_0+n/2-1}\exp\Big\{-\Big(b_0 + \frac{SS_\mu}{2}\Big)\tilde\sigma\Big\}.$$

Answer. $\tilde\sigma \mid x \sim \text{Gamma}(a_n, b_n)$ with $a_n = a_0 + \dfrac{n}{2}, \quad b_n = b_0 + \dfrac{SS_\mu}{2}$.

(c) $\pi(\tilde\sigma \mid x) \propto \tilde\sigma^{-1}\cdot\tilde\sigma^{n/2}e^{-\tilde\sigma SS_\mu/2} = \tilde\sigma^{n/2-1}e^{-\tilde\sigma SS_\mu/2}$.

Answer. $\tilde\sigma \mid x \sim \text{Gamma}\big(\frac{n}{2},\ \frac{SS_\mu}{2}\big)$, which is proper whenever $SS_\mu > 0$. The Jeffreys prior $\tilde\sigma^{-1}$ is the limit of the Gamma(a_0, b_0) kernel $\tilde\sigma^{a_0-1}e^{-b_0\tilde\sigma}$ as $a_0, b_0 \to 0$. Setting $a_0 = b_0 = 0$ in (b) gives exactly this posterior: the Jeffreys analysis is the conjugate analysis with zero prior "observations" and zero prior sum of squares.

(d) $n/2 = 5$ and $SS_\mu/2 = 9$.

Answer. Conjugate Gamma$(2,1)$: $\tilde\sigma \mid x \sim \text{Gamma}(2+5,\ 1+9) = \text{Gamma}(7, 10)$, posterior mean $7/10 = 0.70$.
Jeffreys: $\tilde\sigma \mid x \sim \text{Gamma}(5, 9)$, posterior mean $5/9 \approx 0.556$.
(For comparison, the MLE of $\tilde\sigma$ is $n/SS_\mu = 10/18 = 0.556$, which matches the Jeffreys mean. The Gamma(2,1) prior, with mean 2, pulls the estimate up.)

(e) Integrate out $\tilde\sigma$:

$$\begin{aligned} p(x_f \mid x) &= \int_0^\infty \Big(\frac{\tilde\sigma}{2\pi}\Big)^{1/2} e^{-\frac{\tilde\sigma}{2}(x_f-\mu)^2}\,\frac{b_n^{a_n}}{\Gamma(a_n)}\tilde\sigma^{a_n-1}e^{-b_n\tilde\sigma}\,d\tilde\sigma \propto \Big(b_n + \tfrac{(x_f-\mu)^2}{2}\Big)^{-(a_n+1/2)} \\ &\propto \Big(1 + \frac{1}{2a_n}\frac{(x_f-\mu)^2}{b_n/a_n}\Big)^{-\frac{2a_n+1}{2}}, \end{aligned}$$

which is the kernel of a Student t with $\nu = 2a_n$, location μ and squared scale b_n/a_n.

Answer. $X_f \mid x \sim t_{2a_n}\big(\mu,\ \sqrt{b_n/a_n}\big)$: df $= 2a_n$, location $= \mu$, scale $= \sqrt{b_n/a_n}$ (scale^2 $= b_n/a_n$).
With Gamma$(7, 10)$ from (d): $X_f \mid x \sim t_{14}\big(5,\ \sqrt{10/7}\big) = t_{14}(5,\ 1.195)$, scale$^2 = 1.429$.

Question 4: Normal Model with Unknown Mean and Variance

(a) Multiply the likelihood and the two prior pieces, using $\sum(x_i - \mu)^2 = SS + n(\bar{x} - \mu)^2$:

$$p(\mu, \sigma^2 \mid x) \propto (\sigma^2)^{-n/2} e^{-\frac{SS + n(\mu-\bar{x})^2}{2\sigma^2}} \cdot (\sigma^2)^{-1/2} e^{-\frac{p_0(\mu-\mu_0)^2}{2\sigma^2}} \cdot (\sigma^2)^{-\nu_0/2-1} e^{-\frac{SS_0}{2\sigma^2}}.$$

Complete the square in μ:

$$n(\mu-\bar{x})^2 + p_0(\mu-\mu_0)^2 = (p_0+n)\Big(\mu - \frac{p_0\mu_0 + n\bar{x}}{p_0+n}\Big)^2 + \frac{p_0 n}{p_0+n}(\bar{x}-\mu_0)^2 = p_n(\mu-\mu_n)^2 + \frac{p_0 n}{p_n}(\bar{x}-\mu_0)^2.$$

Substituting and regrouping,

$$p(\mu, \sigma^2 \mid x) \propto \underbrace{(\sigma^2)^{-1/2} \exp\Big\{-\frac{p_n(\mu-\mu_n)^2}{2\sigma^2}\Big\}}_{N(\mu_n,\ \sigma^2/p_n)} \cdot \underbrace{(\sigma^2)^{-\frac{\nu_0+n}{2}-1} \exp\Big\{-\frac{SS_0 + SS + \frac{p_0 n}{p_n}(\bar{x}-\mu_0)^2}{2\sigma^2}\Big\}}_{\mathrm{IG}(\nu_n/2,\ SS_n/2)}.$$

Answer. The posterior is again normal–inverse-gamma: $\mu \mid \sigma^2, x \sim N(\mu_n, \sigma^2/p_n)$ and $\sigma^2 \mid x \sim \mathrm{IG}(\nu_n/2, SS_n/2)$, with $p_n = p_0 + n$, $\mu_n = \frac{p_0\mu_0 + n\bar{x}}{p_0+n}$, $\nu_n = \nu_0 + n$, $SS_n = SS_0 + SS + \frac{p_0 n}{p_0+n}(\bar{x} - \mu_0)^2$.

(b)

Answer. μ_n is a weighted average of the prior mean and the sample mean, with weights proportional to p_0 and n. Since $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/p_0)$ has the same variance as a mean of p_0 observations, p_0 is the *prior sample size*: the prior counts as if we had already seen p_0 observations averaging μ_0. A larger p_0 means a stronger prior. As $p_0 \to 0$, $\mu_n \to \bar{x}$; as $n \to \infty$ with p_0 fixed, the data dominate.

(c)

Answer.

- SS_0: the prior sum of squares, the variability "seen" in the ν_0 prior pseudo-observations (SS_0/ν_0 is a prior guess of σ^2).
- SS: within-sample variability of the data about $\bar{x}$.
- $\frac{p_0 n}{p_0+n}(\bar{x} - \mu_0)^2$: between-source variability, the disagreement between the prior mean and the sample mean. If both are estimates of the same μ, then $\bar{x} - \mu_0$ given σ^2 has variance $\sigma^2(1/p_0 + 1/n) = \sigma^2 \frac{p_0+n}{p_0 n}$, so this term is the squared discrepancy standardized by that variance.

It grows with $(\bar{x} - \mu_0)^2$. When the prior and the data strongly disagree, the most coherent explanation within the model is that σ^2 is larger than either source alone suggests. The posterior then inflates its estimate of σ^2, which widens the uncertainty about μ.

(d)

$$p_n = 4 + 8 = 12, \quad \mu_n = \frac{4(10) + 8(12)}{12} = \frac{136}{12} = 11.333, \quad \nu_n = 6 + 8 = 14,$$

$$SS_n = 30 + 42 + \frac{4 \cdot 8}{12}(12 - 10)^2 = 72 + \frac{32}{12} \cdot 4 = 72 + 10.667 = 82.667.$$

Answer. $p_n = 12,\ \mu_n = 34/3 \approx 11.333,\ \nu_n = 14,\ SS_n = 248/3 \approx 82.667.$
$\mu \mid \sigma^2, x \sim N\big(11.333,\ \sigma^2/12\big), \quad \sigma^2 \mid x \sim \text{IG}\big(7,\ 41.333\big)$ (posterior mean of σ^2: $41.333/6 = 6.889$).

(e) $\sqrt{SS_n/(p_n\nu_n)} = \sqrt{82.667/168} = \sqrt{0.4921} = 0.7015.$

Answer. $\mu \mid x \sim t_{14}\big(11.333,\ 0.7015\big)$ (scale^2 $= 0.492$); a 95% interval is $11.333 \pm t_{14,0.975}(0.7015) = 11.333 \pm 2.145(0.7015) = (9.83,\ 12.84)$.
The marginal is $p(\mu \mid x) = \int N(\mu; \mu_n, \sigma^2/p_n)\,\text{IG}(\sigma^2; \nu_n/2, SS_n/2)\,d\sigma^2$, a scale mixture of normals over an inverse-gamma variance, which integrates to a Student t. Since σ^2 is unknown, we average over many plausible variances instead of plugging in one value. The averaging gives heavier tails than a normal, which reflects the extra uncertainty from estimating σ^2. As $\nu_n \to \infty$ the t approaches the normal.

Question 5: Posterior Simulation and Prior Sensitivity

(a) From the data: $n = 10$, $\bar{x} = 100.7/10 = 10.07$, and $SS = \sum(x_i - \bar{x})^2 = 7.001$.

$$p_n = 13, \quad \mu_n = \frac{3(9) + 10(10.07)}{13} = \frac{127.7}{13} = 9.8231, \quad \nu_n = 14,$$

$$SS_n = 8 + 7.001 + \frac{3 \cdot 10}{13}(10.07 - 9)^2 = 15.001 + 2.3077(1.1449) = 15.001 + 2.642 = 17.643.$$

Answer. $n = 10,\ \bar{x} = 10.07,\ SS = 7.001; \quad p_n = 13,\ \mu_n = 9.823,\ \nu_n = 14,\ SS_n = 17.643.$

(b) R code:

```
set.seed(465)
x <- c(8.4, 10.1, 9.7, 11.2, 10.5, 9.1, 10.8, 11.0, 9.6, 10.3)
n <- length(x); xbar <- mean(x); SS <- sum((x - xbar)^2)

nig_post <- function(mu0, p0, nu0, SS0) {
  pn <- p0 + n
  mun <- (p0 * mu0 + n * xbar) / pn
  nun <- nu0 + n
  SSn <- SS0 + SS + (p0 * n / pn) * (xbar - mu0)^2
  list(pn = pn, mun = mun, nun = nun, SSn = SSn)
}

sim_post <- function(post, S = 50000) {
  # sigma^2 ~ IG(nun/2, SSn/2) <=> 1/sigma^2 ~ Gamma(shape = nun/2, rate = SSn/2)
  sigma2 <- 1 / rgamma(S, shape = post$nun / 2, rate = post$SSn / 2)
  mu <- rnorm(S, mean = post$mun, sd = sqrt(sigma2 / post$pn))
  data.frame(mu = mu, sigma2 = sigma2)
```

```
}

post1 <- nig_post(mu0 = 9, p0 = 3, nu0 = 4, SS0 = 8)
draws1 <- sim_post(post1)

# (c) mu summaries
mean(draws1$mu); mean(draws1$mu > 10); quantile(draws1$mu, c(.025, .975))
# (d) sigma^2 summaries and plots
mean(draws1$sigma2); quantile(draws1$sigma2, c(.025, .975))
par(mfrow = c(1, 2))
plot(density(draws1$mu), main = expression("Posterior of " * mu), xlab = expression(mu))
plot(density(draws1$sigma2), main = expression("Posterior of " * sigma^2),
     xlab = expression(sigma^2), xlim = c(0, 4))

# (e) weak prior
post2 <- nig_post(mu0 = 0, p0 = 0.01, nu0 = 0.01, SS0 = 0.01)
draws2 <- sim_post(post2)
mean(draws2$mu); mean(draws2$mu > 10); quantile(draws2$mu, c(.025, .975))
mean(draws2$sigma2); quantile(draws2$sigma2, c(.025, .975))
```

Monte Carlo estimates below come from 50,000 draws. Your R values will differ in the second or third decimal depending on the seed. The exact values come from $\mu \mid x \sim t_{\nu_n}(\mu_n, \sqrt{SS_n/(p_n\nu_n)})$ and $\sigma^2 \mid x \sim \text{IG}(\nu_n/2, SS_n/2)$, and they are what the simulation should approximate.

(c)

Answer.

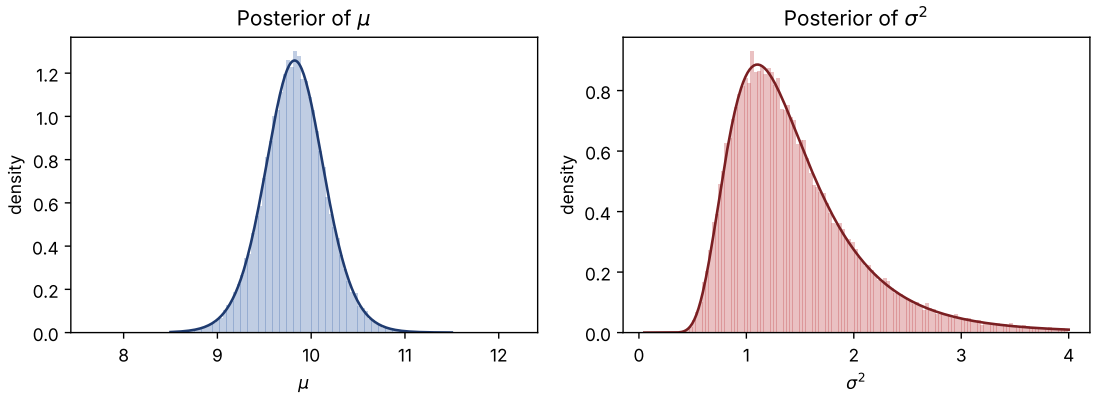
	Monte Carlo	Exact
$E[\mu \mid x]$	9.821	9.823
$P(\mu > 10 \mid x)$	0.287	0.289
95% CI for μ	(9.155, 10.484)	(9.155, 10.491)

(d)

Answer.

	Monte Carlo	Exact
$E[\sigma^2 \mid x]$	1.472	$\frac{SS_n/2}{\nu_n/2-1} = \frac{8.822}{6} = 1.470$
95% CI for σ^2	(0.673, 3.151)	(0.676, 3.135)

The posterior of μ is symmetric (a t_{14}) and centered just below 10. The posterior of σ^2 is right-skewed, as an inverse-gamma is.



Histograms: 50,000 posterior draws. Curves: exact marginal densities.

(e) Weak prior: $p_n = 10.01$, $\mu_n = 100.7/10.01 = 10.060$, $\nu_n = 10.01$, and $SS_n = 0.01 + 7.001 + \frac{0.1}{10.01}(10.07)^2 = 8.024$.

Answer.

	Prior 1 ($\mu_0 = 9, p_0 = 3$)	Weak prior
$E[\mu \mid x]$	9.823	10.060
95% CI for μ	(9.155, 10.491)	(9.430, 10.690)
$P(\mu > 10 \mid x)$	0.289	0.582
$E[\sigma^2 \mid x]$	1.470	1.002
95% CI for σ^2	(0.676, 3.135)	(0.392, 2.467)

(Exact values; Monte Carlo estimates agree to about ± 0.005.)

Under the weak prior the posterior of μ sits essentially at $\bar{x} = 10.07$, as expected when $p_0 \approx 0$. The first prior pulls the mean down by about 0.24 toward $\mu_0 = 9$. With $p_0 = 3$ against $n = 10$, the prior has weight $3/13 \approx 23\%$. The interval widths are similar (about 1.34 vs. 1.26), but the intervals are shifted, and $P(\mu > 10 \mid x)$ roughly doubles from 0.29 to 0.58. The first prior also inflates σ^2: its prior guess $SS_0/\nu_0 = 2$ exceeds the sample variance $7.001/9 \approx 0.78$, and the prior–data conflict term adds 2.64 to SS_n.

Conclusion: with only $n = 10$ observations, the data are not strong enough to wash out the informative prior. Point estimates are in the same neighborhood, but conclusions such as $P(\mu > 10)$ are clearly prior-sensitive. A prior worth 3 observations is substantial relative to 10. With much larger n the two analyses would converge.